Neel Gali

Senior AI Engineer – Generative AI & LLMs

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Industrial AI Platform

- Lead development of production-grade **Generative AI** solutions, utilizing **LangChain** and **OpenAI API** on **AWS SageMaker**, achieving 30% reduction in operational manual review time.
- Architected and implemented **RAG architectures** for complex knowledge retrieval using **vector databases** like **Pinecone** and **Elasticsearch**, boosting query accuracy by 25%.
- Optimized **LLM inference efficiency** via quantization and batching techniques on **AWS EC2** instances, reducing API latency for core services by 40%.
- Designed and deployed **agentic AI systems** with **LangGraph** and custom tools, automating routine industrial support tasks, saving 15 engineering hours monthly.
- Established robust **MLOps workflows** for **LLM deployment** including **CI/CD** pipelines and **model lifecycle management** with **Weights & Biases**, accelerating releases by 20%.
- Implemented **hallucination detection** and **guardrails** for production **LLMs**, ensuring responsible AI practices and improving response reliability by 35%.
- Developed scalable **backend services** and **APIs** in **Python** using **FastAPI** for real-time **AI/ML systems**, supporting over 500 concurrent users effectively.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Logistics Optimization Platform

- Developed and deployed **LLM-powered applications** using **Python** and **Hugging Face Transformers** on **Azure ML**, enhancing logistics routing efficiency by 18%.
- Implemented **RAG systems** for intelligent document processing, leveraging **OpenAI API** and **LlamaIndex** to improve data extraction accuracy by 22%.
- Contributed to **LLM fine-tuning** efforts on custom datasets with **PyTorch** and **AWS SageMaker**, achieving 15% better performance on specific domain tasks.
- Built and maintained **data pipelines** for large-scale ML model training, utilizing **Apache Kafka** and **AWS Lambda**, processing 100GB of daily sensor data.
- Designed and implemented **vector search systems** for semantic search capabilities, integrating with **Elasticsearch** and custom embeddings, reducing query times by 200ms.
- Developed scalable **REST APIs** in **Python** with **FastAPI** for ML model serving, handling 5000 requests per minute with sub-200ms latency.
- Collaborated on establishing **model evaluation** frameworks and **observability** tools like **LangSmith**, improving model performance monitoring and issue resolution by 25%.
- Containerized **AI/ML systems** using **Docker** and orchestrated deployments with **Kubernetes** on **AWS EKS**, ensuring high availability and fault tolerance.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Enterprise SaaS Platform

- Developed robust **backend services** in **Java** for an enterprise **SaaS platform**, enhancing data processing capabilities and improving system uptime by 10%.
- Designed and implemented **REST APIs** using **Spring Boot** for customer-facing features, integrating with **PostgreSQL** to support 50,000 active users.
- Optimized database queries and schemas for **PostgreSQL** and **MongoDB**, reducing report generation time by 30% for critical business intelligence.
- Built scalable **data pipelines** for analytics and reporting, processing 1M daily user events on **AWS S3** and **Lambda**, ensuring data integrity.
- Implemented and maintained **CI/CD pipelines** using **Jenkins** and **Git** for automated testing and deployment, accelerating release cycles by 25%.

Skills

Languages: Python, Java
Frameworks & Tools: PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, FastAPI, Docker, Kubernetes, Git, Apache Kafka, Spring Boot, Jenkins, LangGraph, OpenAI API, LangSmith, Weights & Biases
Cloud & Infra: AWS, Azure, CI/CD, AWS SageMaker, AWS EC2, AWS EKS, AWS Lambda, AWS S3, Azure ML
Databases & Search: PostgreSQL, MongoDB, Elasticsearch, Pinecone
Concepts: Generative AI, LLM Deployment, RAG Architectures, MLOps, System Design, REST APIs, Microservices, Data Pipelines, Model Evaluation, Observability, Prompt Engineering, Hallucination Detection, Responsible AI, Agentic AI Systems, LLM Fine-tuning, Inference Efficiency, Guardrails, Model Lifecycle Management

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India